Astrea Academy Trust
INSPIRING BEYOND MEASURE

Name .....

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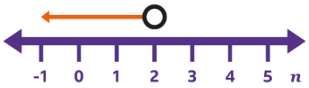
YEAR 11 CORE MATHEMATICS

Unit 1: Further Algebra – Booklet 1/2

Linear Inequalities & Rearranging Formulae

$> \quad < \quad \geq \quad \neq \quad \leq$



$$s = ut + \frac{1}{2}at^2$$

KEY VOCABULARY LIST	
Constant	A fixed number on its own in an algebraic statement.
Variable	A symbol (usually a letter) standing in for an unknown numerical value in an equation.
Equation	A mathematical statement that two things are **equal** e.g.. $1 + 2 = 3$
Inequality	A relationship between two variables that are <u>not equal</u>, using the symbols: $\neq$, $<$, $>$, $\leq$ and $\geq$
Formula	A relationship between multiple variables, expressed algebraically. E.g. $A = bh$
Subject	The single variable that everything else is equal to. The subject should have no coefficient, constant, power or other function attached to it, and appear only once.
Solve	To solve means to find a solution. For an inequality this is a list or expression giving the possible values of the variable. Simultaneous equations have **multiple** solutions.
Balancing	Performing the same mathematical operation on both sides of the equation to maintain equality. Used to solve linear equations by isolating the variable. In solving inequalities this method is ensures the relationship remains consistent.
Simultaneous Equations	Two or more equations that share variables. e.g. $x + y = 6$ $-3x + y = 2$
Elimination Method	"Eliminate" means to remove - this method works by removing variables until there is just one left. This is done by manipulating equations to give variables with equal coefficients then adding/subtracting them to eliminate them.
Substitution Method	"Substitute" means to replace – this method works by writing one of the equations in the form "variable = ..." & substituting this into the other equations, giving an equation in a single variable to be solved.

Unit 1: Further Algebra – Booklet 1/2

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See Year 11 Core Unit 1 - Booklet 2/2 for:	
SUBTOPIC 4A	RECAP: Plotting Linear Graphs
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SUBTOPIC 5A	Simultaneous Equations: Graphical
SUBTOPIC 5B	Simultaneous Equations: Algebraic - By Elimination
SUBTOPIC 5C	Simultaneous Equations – Form & Solve
	GCSE QUESTIONS

BOOKLET STRUCTURE

- **Start of subtopic:**
  - Read and annotate the information in the Frayer model or introduction text
  - I do (teacher led example).
  - We do (teacher and scholar led example).
  - You do (independent scholar example).
  - Whiteboard check.
  - Independent practice and mixed practice.
- **End of subtopic:**
  - Skills checks – key questions to help your teachers to pinpoint any issues.
  - GCSE questions – to ensure you are confident with the style you will encounter in your exams.

Subtopic 1A: Solving Equations (RECAP)

LEARNING OUTCOME: How to solve equations with One Variable

Keywords:
- Equations
- Solving
- Inverse operations
- Substitution
- Variables

Objectives:

1. To understand what an equation is
2. To become fluent in methods for solving equations

1. **Introduction to Solving Equations:**
2. Equations are a mathematical statement that two things are **equal.**
3. You could see this with an example of an equation $1 + 3 = 4$ where the value on the **left-hand side of the equation** is exactly the same as on the **right-hand side of the equation.**

4. Equations in your maths lessons will more usually be in an **algebraic form**, that is they will contain
5. **variables** which represent unknown numbers (in fact, mathematicians sometimes refer to these
6. variables as "unknowns").
7. Generally, our challenge will be to work out the value of the variable by **solving**.

Examples	Non - Examples
$3 \times 4 = 12$	$3x > 2x + 7$
$3x = 2x + 7$	$3x = 2 \times 6$	$3x$
$3 \times 4 = 2 \times 6$	$2 + 7 = 15 - 6$	$3 \times 4 \neq 15$
$2x^2 + 7 = 15$	$15 = 2x + 7$	$2 + 7 \neq 15 - 5$
$2x + 7 = 15$	$2x + 7 = 2x^2 + 7$	$2 + 7 < 15 - 5$

8. The simplest equations can be thought of as 'word problems'. For example, the equation, $x + 7 = 10$
9. can be thought of as "There is an unknown number, and when you add 7 to it, you get 10."
10. This means that the unknown number must be 3. In this case, we have solved the equation when we
11. write $x = 3$.

12. You have probably been solving equations for a long time without really realising it. Look at the
13. examples below:
    a) I think of a number. I double it and get 12. What number was I thinking of?
    b) I think of a number. I subtract 5 from it and I get 15. What number was I thinking of?
    c) I think of a number. I multiply it by 3 and I get 33. What number was I thinking of?

14. These kinds of problems can often be done in our heads using **inverse operations.** This idea can still
15. be used when the equations get more challenging.

Inverse Operations

An inverse operation is the opposite operation that 'undoes' the first operation.

Operation	Inverse Operation
Addition	Subtraction
Subtraction	Addition
Multiplication	Division
Division	Multiplication
Squaring	Square Rooting
Cubing	Cube Root
Holding to a power of x	Xth root

	Example 1. I do…	*Example 2. We do…*	*Example 3. You do…*
Steps	Fill in the gaps:	Fill in the gaps:	Fill in the gaps:
Use your knowledge of operations to work out the relevant inverse operation	"The inverse of adding 7 to a number is _____ _____"	"The inverse of dividing by 9 is _____ _____"	"The inverse of subtracting 8 from a number is _____ _____"
	Example 4. I do…	*Example 5. We do…*	*Example 6. You do…*
	"The inverse of subtracting 4 from a number is _____ _____"	"The inverse of multiplying by 11 is _____ _____"	"The inverse of squaring a number is _____ _____"

Solving One-Step Equations

One-Step Equations: Addition and Subtraction

1. The objective when solving an equation is to work out the value of the variable, so that you are left
2. with a final answer such as $x = 8.5$. As mathematicians we call this **isolating the variable.**

3. **Even if you can work these out in your head, please write down all of the working out, as you will**
4. **apply this technique to more and more difficult questions later!**

	Example 7. I do…	*Example 8. We do…*	*Example 9. You do…*
Steps	Solve:	Solve:	Solve:
1. Decide whether the variable is on the right-hand side (RHS) or left-hand side (LHS) of the equation. 2. Look carefully to decide the current operations happening to the variable 3. Apply the inverse operation to **both sides** of the equation to isolate the variable and solve	$x + 7 = 13$	$3 + x = 30$	$3.7 = 2.5 + x$
	Example 10. I do…	*Example 11. We do…*	*Example 12. You do…*
	$25 = x - 10$	$x - 5.6 = 4.3$	$x - 11 = 22$

	Example 13. I do…	*Example 14. We do*	*Example 15. You do*
Steps	Solve:	Solve:	Solve:
1. Decide whether the variable is on the right-hand side (RHS) or left-hand side (LHS) of the equation.	$2y = 18$	$1.2 = 4a$	$7x = 15$
	Example 16. I do…	*Example 17. We do*	*Example 18. You do*
2. Look carefully to decide the current operations happening to the variable 3. Apply the inverse operation to **both sides** of the equation to isolate the variable and solve	$10 = \frac{y}{5}$	$\frac{b}{3} = 11$	$2 = \frac{z}{4}$

Mixed Practice

Question 1. Solve each of the equations. Show all your working.

a) $x + 1 = 10$	b) $\frac{c}{3} = 8$	c) $28 = 7a$	d) $14 = y - 9$
e) $119 = n + 41$	f) $7p = 50$	g) $24 = \frac{1}{2}b$	h) $x - 11.7 = 11.1$
i) $20 = \frac{h}{13}$	j) $77 = 14a$	k) $2.5 = \frac{v}{10}$	l) $13 = z - 3.8$
m) $\frac{1}{7}m = 40$	n) $12 = c + 15$	o) $1.91 + x = 2.7$	p) $15 = -5f$
q) $15x = 3$	r) $15 = 3x$	s) $\frac{x}{15} = -3$	t) $15 = x - 3$

	Example 19. I do…	*Example 20. We do*	*Example 21. You do*
Steps	Solve:	Solve:	Solve:
1. Decide whether the variable is on the right-hand side (RHS) or left-hand side (LHS) of the equation. 2. Look carefully to decide the current operations happening to the variable 3. Apply the inverse operation to **both sides** of the equation to isolate the variable and solve	$\sqrt{x} = 4$	$9 = \sqrt{y}$	$\sqrt{x} = 7$
	Example 22. I do…	*Example 23. We do*	*Example 24. You do*
	$x^2 = 25$	$36 = y^2$	$x^2 = 144$
	Example 25. I do…	*Example 26. We do*	*Example 27. You do*
	$\sqrt[3]{x} = 3$	$3 = \sqrt[4]{x}$	$\sqrt[3]{x} = -2$
	Example 28. I do…	*Example 29. We do*	*Example 30. You do*
	$x^3 = -125$	$16 = y^4$	$x^3 = 64$

Independent Practice

Question 2. Solve each of the equations. Show all your working.

a) $x^2 = 49$	b) $\sqrt{y} = 3$	c) $a^3 = 125$	d) $4 = \sqrt[3]{c}$
e) $x^2 = \frac{36}{81}$	f) $\sqrt{y} = 37$	g) $10 = \sqrt[4]{c}$	h) $x^3 = 0.008$

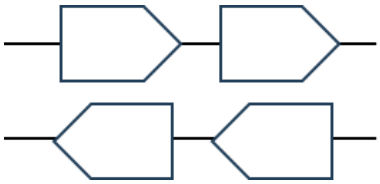
Solving Two-Step Equations

1. It is helpful to think of solving equations as **"working backwards"** – you know something has
2. happened to the variable and you need to work backwards to find its value. This is why we have been
3. using the **inverse operation** to isolate the variable.

4. Often, we will need to do more than one step to solve the equation, so we have to consider the
5. inverse of **multiple operations**. You will need to do the steps in the correct order by considering the
6. **order of operations** to solve successfully!

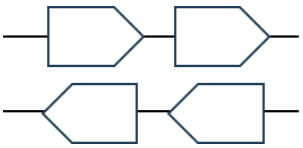
I do…

The function machine below shows one way of thinking about the equation $3x + 5 = 38$

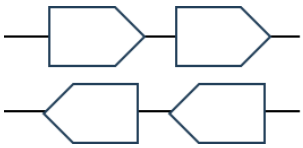
$3x + 5 = 38$		$x =$ ___

When **solving** you are 'working backwards', so work in reverse order with the inverse operations.

We do…

$6x - 2 = 70$		$x =$ ___

You do…

$\frac{x+2}{3} = 5$		$x =$ ___

1. As equations get more complex, this is not always the best representation, but the same kind of
2. thinking can make sure we complete each step in the right order.

3. Instead, we manipulate the equations themselves, showing the inverse operations being applied to
4. each side of the "=" to **keep it balanced.**

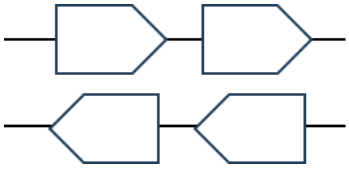
	Example 1. I do…	***Example 2. We do…***	***Example 3. You do…***
Steps	Solve:	Solve:	Solve:
1. Look carefully to decide the current operations happening to the variable. (Use a function machine if it helps) 2. Apply the inverse operations to **both sides** of the equation ('reverse BIDMAS' may help with this!)	$3x + 5 = 38$	$6x - 2 = 70$	$\frac{x+2}{3} = 5$

Two-step Equations with Multiplication

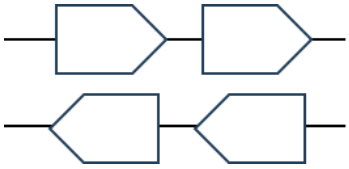
	Example 4. I do…	*Example 5. We do…*	*Example 6. You do…*
Steps	Solve:	Solve:	Solve:
1. Look carefully to decide the current operations happening to the variable. 2. Apply the inverse operations to **both sides** of the equation ('reverse order of operations' may help with this!)	$13 = 4x + 9$	$53 = 10y - 7$	$7 = 2 + 3b$

Two-step Equations with Brackets

Equations **involving brackets** can be solved using the order of operations.

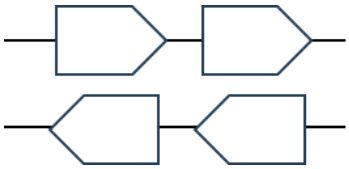
$3(x + 5) = 36$		$x =$ ___

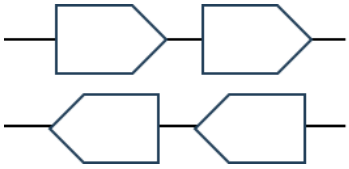
Or we can expand the brackets and solve as above.

$3(x + 5) = 36$ expands to $3x + 15 = 36$		$x =$ ___

	Example 7. I do…	*Example 8. We do…*	*Example 9. You do…*
Steps	Solve:	Solve:	Solve:
1. Decide whether the variable is on the right-hand side (RHS) or left-hand side (LHS) of the equation. 2. Look carefully to decide the current operations happening to the variable. 3. Apply the inverse operations to **both sides** of the equation ('reverse order of operations' may help with this!)	$4(x + 3) = 44$	$15 = 2(n - 7)$	$13 = 3(4 + a)$

1. Equations **with divisions** also need careful consideration for the order of operations.
2. These two equations **look very similar**, but the division occurs at a different stage for each!
3. This affects the order of the inverse operations used to isolate the variable.

$\frac{x}{3} + 1 = 5$		$x =$ ___

$\frac{x+1}{3} = 5$		$x =$ ___

	Example 10. I do…	***Example 11. We do…***	***Example 12. You do…***
Steps	Solve:	Solve:	Solve:
1. Look carefully to decide the current operations happening to the variable. 2. Apply the inverse operations to **both sides** of the equation ('reverse order of operations' may help with this!)	$\frac{x+4}{3} = 10$	$12 = \frac{x}{5} - 6$	$\frac{x-12}{4} = 15$
	Example 13. I do…	***Example 14. We do…***	***Example 15. You do…***
	$\frac{x}{3} + 4 = 10$	$12 = \frac{x-6}{5}$	$15 = \frac{x}{4} - 3$

Mixed Practice

Question 3. Solve each of the equations. Show all your working.

a) $7 + 3x = 22$	b) $300 = 40z - 100$	c) $\frac{a}{3} + 5 = 12$	d) $6(n + 3) = 42$
e) $7 = \frac{y-10}{4}$	f) $-14 = 13 + 3x$	g) $3b + 11 = 61$	h) $5(4 + x) = 24$
i) $7 = 6 + 25x$	j) $\frac{y+32}{5} = -4$	k) $3.2 = \frac{h}{6} - 5.1$	l) $4z - 1 = -100$

Challenge – More than Two Steps

Question 4. Solve each of the equations. Show all your working.

a) $4(5x - 3) = 100$	b) $\frac{7p}{2} - 12 = 30$	c) $-7 = \frac{y+5}{6} - 4$	d) $\frac{3(2x-4)}{10} = 2.7$

Two-step Equations with Powers

	Example 16. I do…	*Example 17. We do…*	*Ex18. You do…*
Steps	Solve:	Solve:	Solve:
1. Look carefully to decide the current operations happening to the variable. 2. Apply the inverse operations to **both sides** of the equation.	$e^2 + 6 = 22$	$21 = x^3 - 6$	$4 + a^2 = 21$

Two-step Equations with Roots

	Example 19. I do…	*Example 20. We do…*	*Example 21. You do…*
Steps	Solve:	Solve:	Solve:
1. Look carefully to decide the current operations happening to the variable. 2. Apply the inverse operations to **both sides** of the equation in the correct order.	$\sqrt{x} + 8 = 10$	$\sqrt[3]{f} + 5 = 2$	$10 = \sqrt{s} + 2$

Independent Practice

Question 5. Solve each of the equations. Show all your working.

a) $\sqrt{x} + 7 = 10$	b) $t^2 + 2 = 11$	c) $\sqrt{s} + 9 = 10$	d) $y^2 - 5 = -7$
e) $\sqrt[3]{f} + 1 = 7$	f) $y^3 - 10 = -18$	g) $\sqrt{x} + 8 = 100$	h) $t^2 + 1 = \frac{17}{16}$

Freya thinks she has solved the equation below. She is **incorrect**. Can you explain why?

$$10 - x = 4$$
$$-10 \qquad\qquad -10$$
$$x = -6$$

1. Where there are **negative coefficients** of a variable, we need to be careful to **isolate the positive**
2. **variable**. The easiest & most reliable way to do so is to **add the positive equivalent to each side** to
3. create an equation more like those we have solved so far.

	Example 22. I do…	***Example 23. We do…***	***Example 24. You do…***
Steps	Solve:	Solve:	Solve:
1. Where the coefficient of x is negative, we need to add this to both sides of the equation to give 'a positive x'	$12 - x = 7$	$15 = 12 - x$	$-x - 3 = 7$
2. Solve as normal!	***Example 25. I do…***	***Example 26. We do…***	***Example 27. You do…***
	$12 - 2x = 7$	$15 = -6x + 12$	$5(3 - x) = 20$

Independent Practice

Question 6. Solve each of the equations. Show all your working.

a) $24 - x = 17$	b) $-n + 4.3 = 3.1$	c) $3 = -10 - y$	d) $31 - 3k = 4$
e) $13 = -8 - 3a$	f) $-\frac{x}{2} = 12.3$	g) $100 - x^2 = 36$	h) $\frac{50-x}{7} = 5$

Question 1

(a) Solve $b - 7 = 12$

$b =$ .....

(1)

(b) Solve $5e = 40$

$e =$ .....

(1)

(c) Solve $4m + 6 = 15$

$m =$ .....

(2)

Question 2

Solve $4(x - 6) = 44$

$x =$ .....

(Total for question = 2 marks)

Question 3

(a) Solve $3m = 36$

$m =$ .....

(1)

(b) Solve $7 - x = 3$

$x =$ .....

(1)

Question 4

$x - 1 = 2$

Work out the value of $2x^2$

.....

(Total for question = 3 marks)

Question 5

(a) Solve $x + x + x = 6$

$x =$ .....

(1)

(b) Solve $t + 5 = 20$

$t =$ .....

(1)

(c) Solve $4y = 36$

$y =$ .....

(1)

(d) Solve $\frac{1}{2}f + 5 = 12$

$f =$ .....

(2)

Question 6

Solve $5 = \frac{100}{x}$

.....

(Total for question = 1 mark)

Question 7

(a) Expand $y(y + 5)$

.....

(1)

(b) Factorise $4a - 6$

.....

(1)

(c) Solve $2(5x - 4) = 21$

$x =$ .....

(3)

(d) Simplify $4e^2f \times 5ef^3$

.....

(1)

Subtopic 1B: Equations with Unknowns on Both Sides

1. Sometimes an equation will not be equal to a given constant, but instead **equal to another**
2. **expression.**
3. We still want to **isolate the variable** to create an equation more like those we have solved previously,
4. but we also need to **eliminate** it on one side of the equation to give the value of the variable.

	Example 28. I do…	***Example 29. We do…***	***Example 30. You do…***
Steps	Solve:	Solve:	Solve:
1. Find the variable that is repeated on each side of the equation. 2. Use the inverse operation of one of these terms to 'eliminate' it from one side of the equation. (Carefully consider the steps here and **try to keep the coefficient positive**) 3. Solve as normal!	$5x = 2x + 18$	$5x - 11 = 7x$	$16 + 2x = -6x$
	Example 31. I do…	***Example 32. We do…***	***Example 33. You do…***
	$25 - 2x = 3x$	$3x = 42 - x$	$16 - 2x = -6x$

Does it matter which 'x' term I eliminate?
For each of the below, describe the most efficient first step.

Equation	First Step	Equation	First Step
$7x - 6 = 5x$		$6 - 7x = 5x$	
$5x = 7x - 6$		$6 - 7x = -5x$	
$5x + 6 = 7x$		$6 - 4x = -5x$	
$6 + 5x = 7x$		$6 + 4x = -5x$	
$6 - 5x = 7x$		$6 + 7x = -5x$	

	Example 34. I do…	*Example 35. We do…*	*Example 36. You do…*
Steps	Solve:	Solve:	Solve:
1. Find the variable that is repeated on each side of the equation. 2. Use the inverse operation of one of these terms to 'eliminate' it from one side of the equation. 3. Solve as normal!	$4x + 1 = 2x + 7$	$6x + 3 = 2x - 13$	$3x - 1 = 23 - x$

Does it matter which 'x' term I eliminate?
For each of the below, describe the most efficient first step.

Equation	First Step	Equation	First Step
$6x + 15 = 9x - 3$		$15 - 6x = 9x - 3$	
$9x + 15 = 6x - 3$		$15 - 6x = 3 + 9x$	
$6x - 15 = 9x - 3$		$15 - 6x = 3 - 9x$	
$6x - 15 = 3 - 9x$		$15 - 6x = -3 - 9x$	

Equations with Unknowns on Both Sides & Brackets

	Example 37. I do…	*Example 38. We do…*	*Example 39. You do…*
Steps	Solve:	Solve:	Solve:
1. Expand the bracket(s) 2. Use the inverse operation of one of these terms to 'eliminate' the repeated variable from one side of the equation. 3. Solve as normal!	$4(x + 1) = 2x + 10$	$3(2x + 4) = 2(x - 6)$	$9(2x - 5) = 3(4x + 7)$

Independent Practice

Question 7. Solve each of the equations. Show all your working.

a) $6x = x + 10$	b) $5x = 7x + 12$	c) $5x - 32 = 9x$	d) $-6x = 2x + 50$
e) $6x + 5 = 10 + 4x$	f) $5x + 14 = 8x - 1$	g) $6 + 2x = 11 - 3x$	h) $11 - x = 7 - 2x$
i) $5(x + 3) = 2x$	j) $3(10 - 2x) = 4x + 1$	k) $5(x + 3) = 3(x + 9)$	l) $2(3 - 4x) = 5(3 - x)$

Question 1

Solve $4x + 5 = x + 26$

$x =$ .....

(Total for question = 2 marks)

Question 2

Solve $5x - 14 = 52 - x$

$x =$ .....

(Total for question = 3 marks)

Question 3

Solve $5x - 6 = 3(x - 1)$

$x =$ .....

(Total for question = 3 marks)

Question 4

Solve $4(x + 3) = 2(x + 4)$

$x =$ .....

(Total for question = 3 marks)

Question 5

Solve $\frac{5 - x}{2} = 2x - 7$

$x =$.....

(Total for question = 3 marks)

Subtopic 2A: Understanding Inequality Notation

LEARNING OUTCOME: To be able to read and write algebraic statements of inequality.

Objectives:

1. To be able to read and write single-sided statements of inequality
2. To be able to read and write double-sided statements of inequality

Keywords:
- Inequality
- Greater than
- Less than
- Variable

Introduction:

1. Equations show when two values are equal in value, using the = symbol.
2. Inequalities, on the other hand, show when values (numbers or expressions) are not equal — when
3. one is greater, smaller, or *possibly* equal to another.
4. In earlier years, you learned how to compare numbers using inequality symbols like < and >
5. e.g. $5 < 10$ or $3 > -4$.
6. Now we will build on that knowledge by using algebra to write and understand inequalities involving
7. unknown values. You'll learn how to read and write single and double-sided inequality statements to describe ranges of possible values, and how to list the integers that satisfy them.

$\neq$	$<$	$>$	$\leq$	$\geq$
Not equal to	Less than	Greater than	Less than <u>or equal to</u>	Greater than <u>or equal to</u>

Understanding Single-Sided Inequalities

Example 1 – I do…	***Example 2 - We do…***	***Example 3 - You do…***
Write as an algebraic inequality:		
x is greater than 6	x is less than or equal to 6.8	6 is less than x

Example 4 – I do…	***Example 5 - We do…***	***Example 6 - You do…***
Describe the algebraic inequality in words:		
$x < 5.2$	$3 > x$	$x \geq -1$

Example 7 – I do…	***Example 8 - We do…***	***Example 9 - You do…***
List five greatest of least possible integer values of x if…		
$x < 5.2$	$3 > x$	$x \geq -1$

Understanding Double-Sided Inequalities

1. A **single-sided inequality** compares a variable to one value, like $\boldsymbol{x > 3}$, meaning x is greater than 3.
2. A **double-sided inequality** sets both a lower and upper limit, like $\boldsymbol{2 < x < 7}$.
3. Technically, this is read as "2 is less than x, which is less than 7". However, it is much easier to think of it as "*x is greater than 2 but less than 7*" or "*x is between 2 and 7*".
4. We must also consider if the inequalities are <u>in</u>clusive or <u>ex</u>clusive – whether they can or cannot be
5. equal to the given values.
6. **Inclusive** inequalities are $\leq$ and $\geq$. The variable **can** take these values.
7. **Exclusive** inequalities are < and >. The variable **cannot** take these values.
8. e.g. $2 \leq x < 7$ would mean "*x is between 2 and 7, <u>inclusive of 2</u>*".

Example 10 – I do…	***Example 11 - We do…***	***Example 12 - You do…***
Write as an algebraic inequality:		
x is between 3 and 15, exclusive of 3 and inclusive of 15	x is less than or equal to 4, and greater than or equal to -7	x has a maximum of 100 and a minimum of 45, inclusive of both

Example 13 – I do…	***Example 14 - We do…***	***Example 15 - You do…***
Describe the algebraic inequality in words:		
$-2 < x \leq 5$	$16 < x < 30$	$-12 \leq x < -1$

Example 16 – I do…	***Example 17 - We do…***	***Example 18 - You do…***
List all of the possible integer values of x if…		
$-2 < x \leq 5$	$16 < x < 30$	$-12 \leq x < -1$

Independent Practice

Question 1. Write as an algebraic inequality:

x is greater than 8	y is less than or equal to 1	-1 is less than z	a is greater than or equal to b
c is greater than 0 but less than or equal to 4	n is between 5 and 15, inclusive of 5 and 15	t is less than 8, but greater than or equal to -1	h has a minimum of -150 (exclusive) and a maximum of 630 (inclusive)

Question 2. Describe the algebraic inequality in words:

$x \leq 5$	$a \succ -2$	$b \geq 54$
$-6 < n < 5$	$300 \leq y < 350$	$-100 < k \leq -42$

Question 3. Write as an algebraic inequality:

a) The cost, c, of a TV is less than £550	
b) To go on a rollercoaster, the height of a person, h, must be over 140cm.	
c) A house is subject to tax if its value, v, is £100,000 or more.	
d) To ride a fairground ride, a child's height, n, must be at least 1.2m but less than 1.5m.	
e) In Europe the temperature, t, of a freezer must be no less than -24°C and no more than -18 °C.	
f) A park does not allow entry for children aged under 4, or for those aged 11 or more. (Where a is the age of a child)	

Question 4. List the possible integer values of x if…

$x \leq 9$	$-5 < x \leq 4$	$-12 < x < -1$

Question 5.

a) Give the maximum possible integer value of x if $x < 45$	b) Give the maximum possible integer value of n if $-2 < n \leq 7$	b) Give the minimum possible integer value of n if $-2 < n \leq 7$

Subtopic 2B - Representing Inequalities on Number Lines

LEARNING OUTCOME: To be able to represent inequalities on number lines

Objectives:

1. To be able to represent single inequalities on a number line
2. To be able to represent double inequalities on a number line

Keywords:
- Inequality
- Greater than
- Less than
- Variable

Introduction:

1. Up to now, we've looked at inequalities by listing the **specific integer values** a variable can take—like
2. ($x = 1, 2, 3$). These are **discrete values**, meaning they're separate and countable.
3. But inequalities can also describe a variable that takes **any value within a range**, including fractions
4. and decimals. This is called a **continuous scale**.
5. We can visualise this **on a number line** using lines/connectors, arrows and open or closed circles and
6. shading.
7. Being able to visualise solutions like this will help you understand all of the possible values make the
8. inequality true.

○	Open Circle	Exclusive	$<$ and $>$
●	Closed Circle	Inclusive	$\leq$ and $\geq$

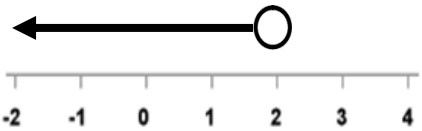
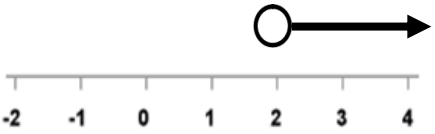
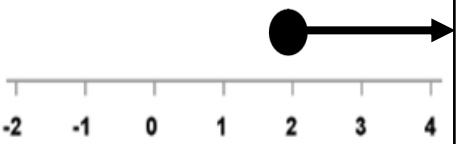
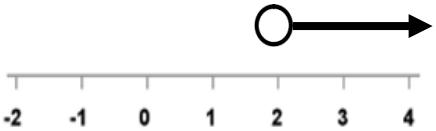
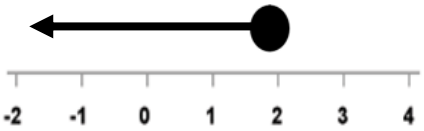
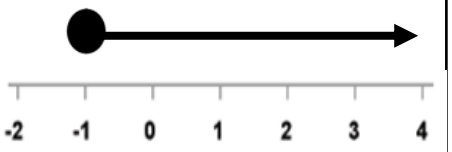
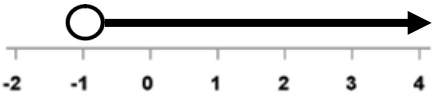
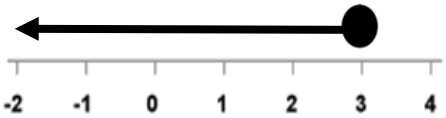
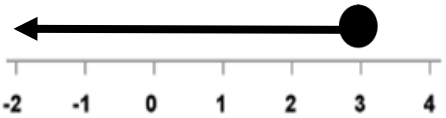
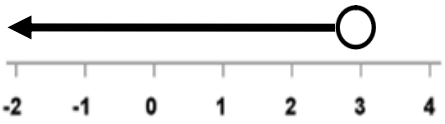
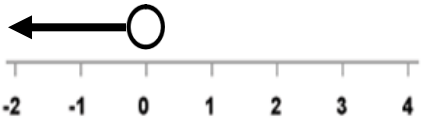
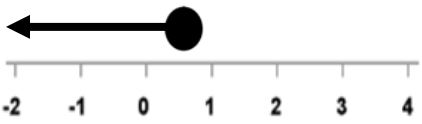
Representing Single-Ended Inequalities on a Number Line

Steps	Represent the inequality on the number line.
1) Identify the number in the inequality. 2) Draw a circle at this number on the number line. 3) Identify whether the unknown is greater than or less than the number. If it is less than, draw an arrow pointing left from the circle. If it is greater than, draw an arrow pointing right from the circle.	***Example 19 – I do…***
	$x < 5$
	Example 20 - We do…
	$x > 3$
	Example 21 - You do…
	$x > -1$

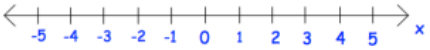
Steps	Represent the inequality on the number line.
1) Identify the number in the inequality. 2) Draw a circle at this number on the number line. 3) Identify whether the unknown is greater than or less than the number. If it is less than, draw an arrow pointing left from the circle. If it is greater than, draw an arrow pointing right from the circle.	***Example 22 – I do…***
	$x \leq 7$
	Example 23 - We do…
	$x \geq 0$
	Example 24 - You do…
	$x \geq -2$

Independent Practice

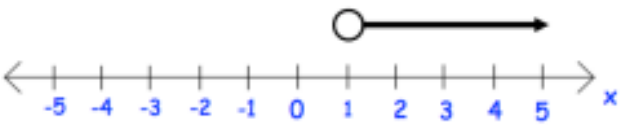
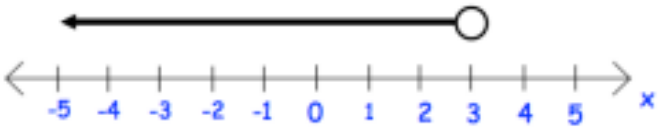
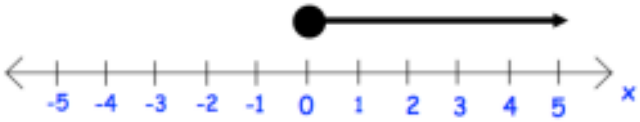
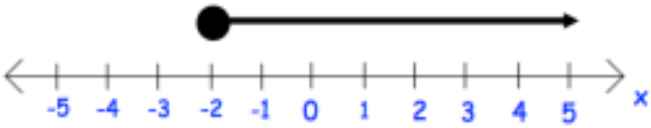
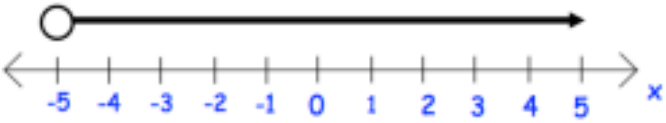
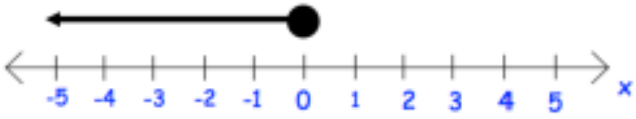
Question 5. True or false? Write T or F in each box.

a) $x \geq 2$ 	b) $x \geq 2$ 	c) $x \geq 2$ 
d) $x > 2$ 	e) $x \leq 2$ 	f) $x \leq -1$ 
g) $x \geq -1$ 	h) $x > 3$ 	i) $x \geq 3$ 
j) $x < 3$ 	k) $x < 0$ 	l) $x \leq 0.5$ 

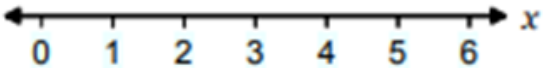
Question 6. Represent the following inequalities on number lines:

a) $x > 2$	b) $x < 4$	c) $x \geq 3$
d) $x \leq 5$	e) $x \geq 0$	f) $x \leq -1$
g) $x < -4$	h) $x > -3$	i) $x \geq -2$
j) $x > 4$	k) $x < -2$	l) $x > -5$ 

Question 7. Write down the inequalities shown below:

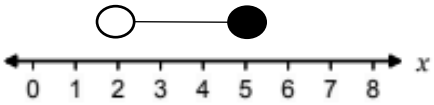
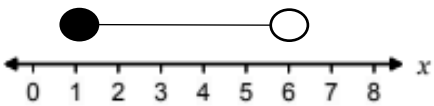
a) 	b) 
c)	d) 
e) 	f)
g) 	h) 

Representing Double-Ended Inequalities on a Number Line

Steps	Represent the inequality on the number line.
1) Identify the numbers in the inequality. 2) Draw circles at each of these numbers on the number line. Remember: greater/ less than is an open circle and greater than or equal to/ less than or equal to is a coloured in circle. 3) Draw a line between the two circles.	***Example 25 – I do…***
	$2 \leq x < 5$
	Example 26 - We do…
	$-1 < x < 1$
	Example 27 - You do…
	$1 \leq x < 3$ 

Independent Practice

Question 8. Fill in the correct inequality symbols to match the diagrams.

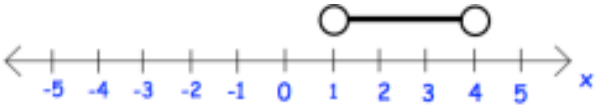
a) $2 \; \square \; x \; \square \; 5$ 	b) $1 \; \square \; x \; \square \; 6$ 

Question 9. Show each inequality on the number line:

a) $1 < x < 4$ 	b) $3 \leq x \leq 5$
c) $2 \leq x < 4$ 	d) $0 < x \leq 3$
e) $-1 < x \leq 3$ 	f) $-2 \leq x < 1$

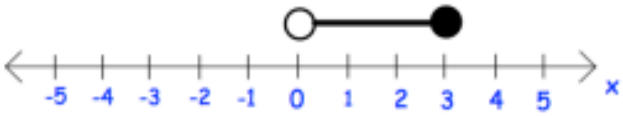
Question 10. Write down the inequalities shown below:

a)

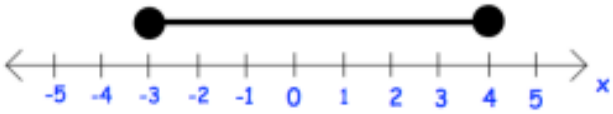


b)

c)

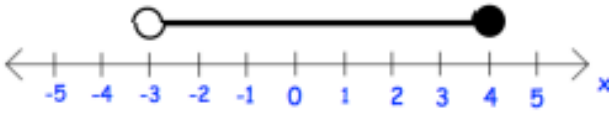


d)



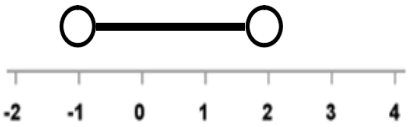
e)

f)

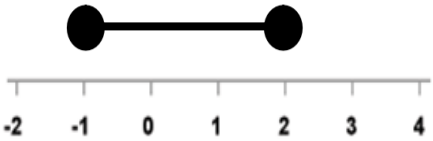


Question 11. True or false? Write T or F in each box.

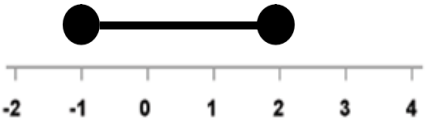
a) $-1 \leq x \leq 2$



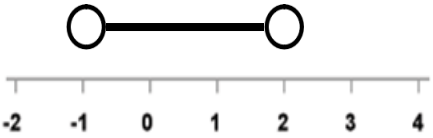
b) $-1 \leq x \leq 2$



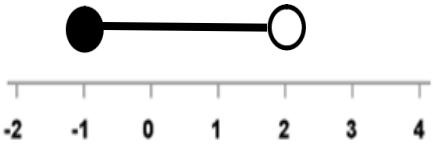
c) $-1 < x < 2$



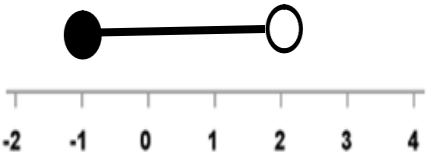
d) $-1 < x < 2$



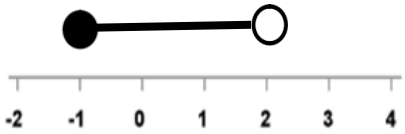
e) $-1 \leq x \leq 2$



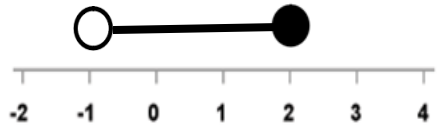
f) $-1 < x \leq 2$



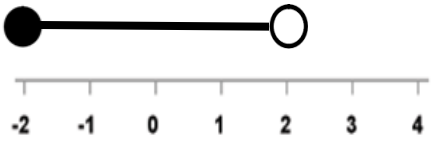
g) $-1 \leq x < 2$



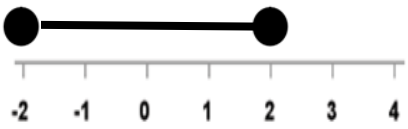
h) $-1 \leq x < 2$



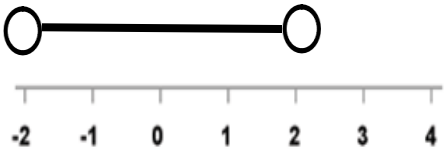
i) $-2 < x \leq 2$



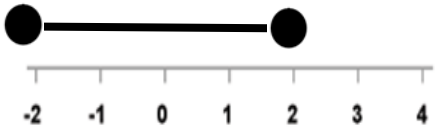
j) $-2 \leq x \leq 2$



k) $-2 < x < 2$



l) $-2 \leq x < 2$



Skills Check

A car rental sign says:
"You must be at least 25 years old to rent a car."

Which inequality represents this information?

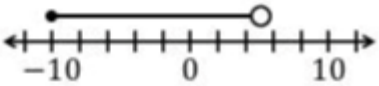
A $a \geq 25$

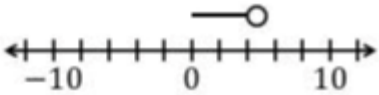
B $a > 25$

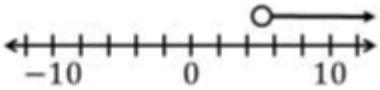
C $a < 25$

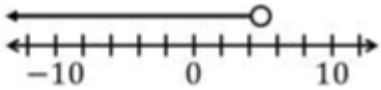
D $a \leq 25$

Which number line best represents the inequality $x < 5$?

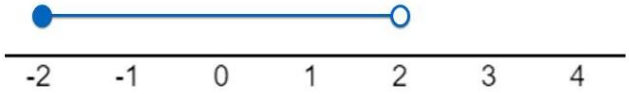
A


B


C


D


Which inequality is shown on this number line?



A $-2 \leq n < 2$

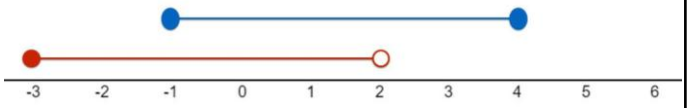
B $-2 \leq n \leq 2$

C $-2 \geq n > 2$

D $-2 \geq n \leq 2$

The inequalities $-1 \leq x \leq 4$ and $-3 \leq x < 2$ are shown on the number line below.

Which list of numbers are value for **both** inequalities?



A $-1, 0, 1$

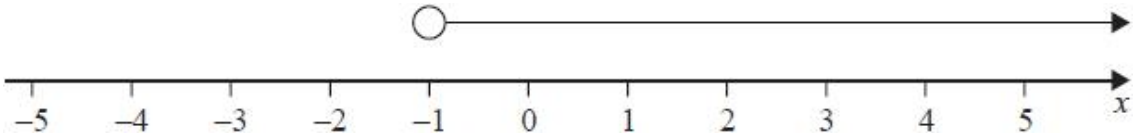
B $0, 1, 2$

C $-1, 0, 1, 2$

D $-3, -2, -1, 0, 1, 2, 3, 4$

Question 1.

(a) Write down the inequality shown on this number line.



.....

(1)

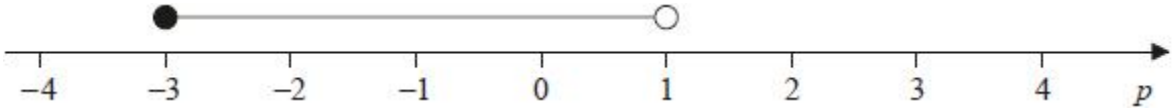
(b) On the number line below, show the inequality $-3 \leq y < 4$

(2)

(Total for question is 3 marks)

Question 2.

Here is a number line.



Write down the inequality shown on the number line

.....

(Total for question is 2 marks)

Question 3.

$$-4 < n \leq 1$$

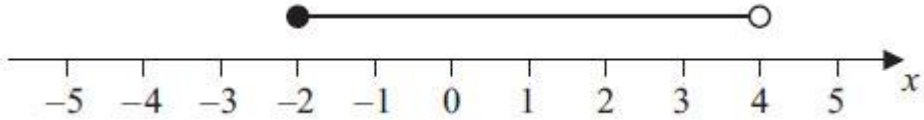
n is an integer.

(a) Write down all the possible values of n.

.....

(2)

(b) Write down the inequalities represented on the number line.



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(2)

(Total for Question is 4 marks)

Subtopic 2C – Solving Single-Ended Linear Inequalities

LEARNING OUTCOME: To understand and solve simple inequalities

Objectives:

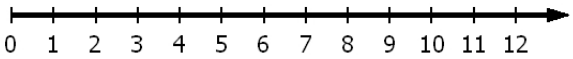
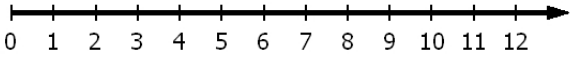
1. To be able to solve one-step inequalities.
2. To be able to solve two-step inequalities
3. To be able to solve inequalities with negative unknowns

Keywords:
- Inequality
- Solution
- Unknown
- Variable

Introduction:

1. Now that you know how to represent inequalities on a number line, it's time to combine this with
2. solving them. You'll begin by recapping how to solve linear inequalities, using the same inverse
3. operation method you've used to solve equations. Once you've solved the inequality, you'll represent
4. the full solution as a lost of integers or on a number line to show all the values that make it true. This step is important because it helps you visualise the solution and check that it makes sense in context.

Solving Single-Ended Linear Inequalities – One Step

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	***Example 1 – I do…***	
	$x + 6 \leq 10$	Represent the solution on the number line below: 
	Example 2 - We do…	
	$\frac{x}{3} > 12$	List the smallest five integers that satisfy this inequality.
	Example 3 - You do…	
	$x + 4 \geq 14$	Represent the solution on the number line below: 

Independent Practice

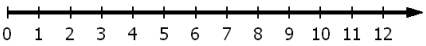
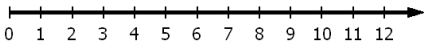
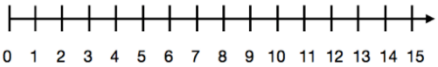
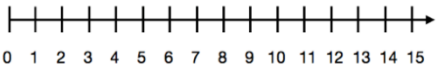
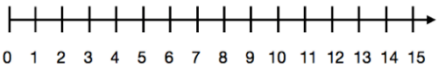
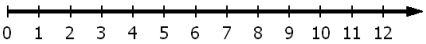
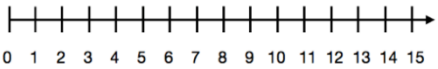
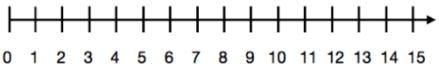
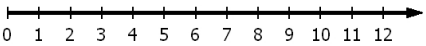
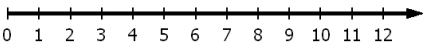
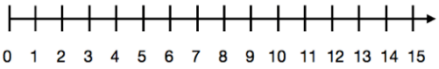
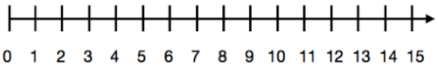
Question 1. Solve the inequalities below:

a) $x + 7 \geq 13$	b) $x + 5 < 7$	c) $x - 10 \leq 3$	d) $x - 3 > 4$
e) $\frac{x}{5} < 10$	f) $4x \geq 24$	g) $\frac{x}{9} \geq 4$	h) $9x < 108$
i) $x - 5 \geq 9$	j) $\frac{x}{8} < 15$	k) $3x > 25.2$	l) $x + 1.5 < 7.8$

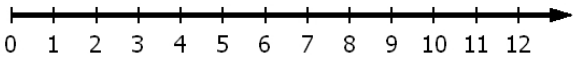
Question 2. Solve the inequalities below & list the five smallest or largest integers that satisfy them:

a) $x + 9 \leq 20$	b) $x - 8 \leq 20$	c) $\frac{x}{4} > 3$	d) $9x > 45$
e) $x + 13 \geq 24$	f) $8x \leq 72$	g) $x - 9.3 > 10.12$	h) $\frac{x}{4} > 1.4$

Question 3. Solve the inequalities, then **represent the solutions on the number lines below**:

a) $x + 4 \geq 15$	b) $x + 2 < 7$	c) $x + 1 \leq 14$
		
d) $x - 4 > 9$	e) $x - 6 \geq 8$	f) $x - 4 \leq 5$
		
g) $\frac{x}{3} > 4$	h) $\frac{x}{2} < 6$	i) $\frac{x}{3} \geq 4$
		
l) $6x \geq 24$	k) $2x > 28$	l) $4x \leq 56$
		

Solving Single-Ended Linear Inequalities – Two Step

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	***Example 4 – I do…***	
	$3x - 5 \geq 22$	Represent the solution on the number line below: 
	Example 5 - We do…	
	$\frac{x}{3} + 1 > 13$	List the smallest five integers that satisfy this inequality.
	Example 6 - You do…	
	$2x + 6 \leq 16$	List the largest five integers that satisfy this inequality.

Independent Practice

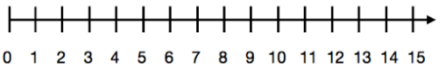
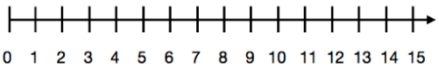
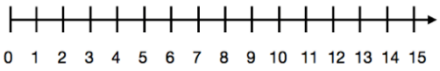
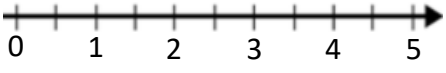
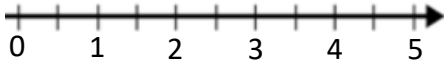
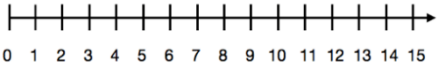
Question 4. Solve the inequalities below:

a) $3x + 7 \geq 13$	b) $2x + 5 < 11$	c) $5x + 9 \leq 44$	d) $2x + 12 \geq 24$
e) $7x - 3 > 4$	f) $4x - 5 \geq 23$	g) $2x - 10 \leq 12$	h) $5x - 8 \leq 17$
i) $\frac{x}{4} + 1 > 7$	j) $\frac{x}{5} + 7 < 11$	k) $\frac{x}{9} - 3 \geq 5$	l) $\frac{x}{8} - 2 < 10$

Question 5. Solve the inequalities, then list the first five smallest or largest integers that satisfy them:

a) $3x + 4 \geq 34$	b) $2x + 2 < 12$	c) $4x + 1 \leq 33$
d) $5x - 4 > 56$	e) $3x - 6 \geq 18$	f) $5x - 11 \leq 89$

Question 6. Solve the inequalities, then **represent the solutions on the number lines below**:

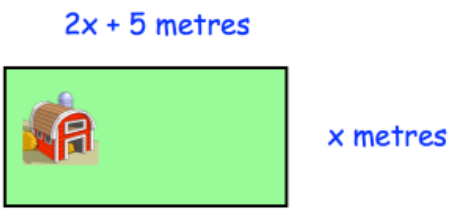
g) $\frac{x}{3} + 6 < 11$	h) $\frac{x}{2} - 3 < 4$	i) $\frac{x}{2} - 5 \geq 2$
		
l) $6x + 1 \geq 4$	k) $2x - 10 > -7$	l) $4x - 3 \leq 57$
		

Question 7.
Lauren goes shopping and has £50 to spend.
She bought a T-shirt and 3 pairs of leggings.
The T-shirt cost £23.
Each pair of leggings cost £x.

a) Form an inequality in terms of x.

b) Solve the inequality to find the highest possible price of the leggings.

Question 8. Farmer Taylor is placing a fence around his field.
He has 300 metres of fencing, but this is not enough.



a) Form an inequality in terms of x.

b) Solve the inequality to find the possible width of the field, x.

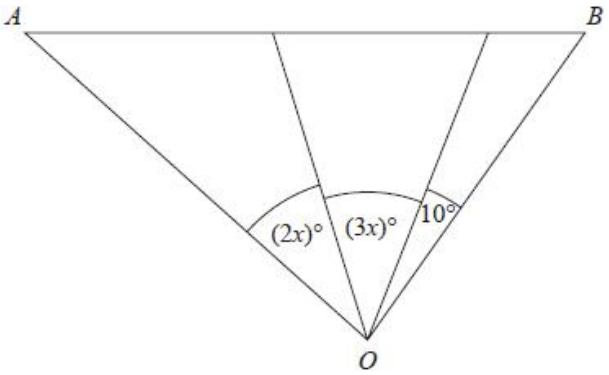
c) What is the smallest possible integer width & length of the field?

Exam Question

The diagram shows triangle *AOB*.

Angle *AOB* is **not** an obtuse angle.

Find the greatest value of x.
You must show all your working.



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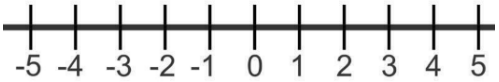
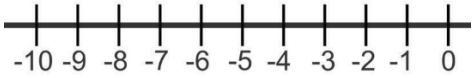
(Total for question = 3 marks)

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	***Example 7 – I do…***	
	$\frac{x}{4} + 10 < 8$	List the largest five integers that satisfy this inequality.
	Example 8 - We do…	
	$2x - 5 > -11$	Represent the solution on the number line below: 
	Example 9 - You do…	
	$\frac{x + 12}{5} \leq 2$	Represent the solution on the number line below: 

Independent Practice

Question 9. Solve the inequalities below:

a) $\frac{x}{5} + 2 > 1$	b) $\frac{x+5}{4} \geq 3$	c) $\frac{x-2}{7} \geq -5$	d) $\frac{x}{8} - 3 < -5$
e)) $3x + 8 \leq 2$	f) $4x + 10 > 6$	g) $4x + 6 < -10$	h) $5x - 3 > -13$

Question 10. Solve the inequalities, then list the first five smallest or largest integers that satisfy them:

a) $3x + 4 \geq 1$	b) $2x + 2 < -8$	c) $4x + 1 \leq -11$
d) $5x - 4 > -14$	e) $3x - 6 \geq -18$	f) $5x - 11 \leq -31$
g) $\frac{x}{3} + 6 > 0$	h) $\frac{x}{2} + 4 < 3$	i) $\frac{x}{3} + 7 \geq 1$
l) $6x + 1 \geq -59$	k) $2x - 10 > -11$	l) $4x - 3 \leq -3.4$

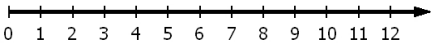
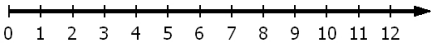
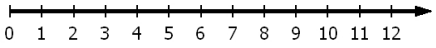
Solving Single-Ended Inequalities with Brackets

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	***Example 10 – I do…***	
	$3(2x+1) > 45$	Represent the solution on the number line below: 0 1 2 3 4 5 6 7 8 9 10 11 12
	Example 11 - We do…	
	$9(2x-3) \leq 80$	List the largest five integers that satisfy this inequality.
	Example 12 - You do…	
	$7(3x+2) \geq 56$	Represent the solution on the number line below: 0 1 2 3 4 5 6 7 8 9 10 11 12

Independent Practice

Question 11. Solve and **represent the solutions on the number lines below:**

a) $4(x+5) > 28$	b) $3(x-2) < 12$	c) $5(x-1) > 25$
0 1 2 3 4 5 6 7 8 9 10 11 12	0 1 2 3 4 5 6 7 8 9 10 11 12	0 1 2 3 4 5 6 7 8 9 10 11 12

d) $7(x+2) \geq 77$	e) $3(2x+1) \leq 27$	f) $2(4x-1) \geq 22$
		

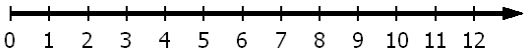
Question 12. Solve the inequalities, then list the first five smallest or largest integers that satisfy them:

a) $5(2x+3) > 65$	b) $10(2x-3) > 100$	c) $3(7x+4) \geq 158$
d) $6(2x+3) \leq 6$	e) $3(2x+10) > 20$	f) $6(3x+11) \leq 12$

Question 13. Find the range of values of x that satisfies **both**

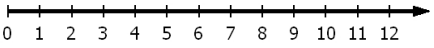
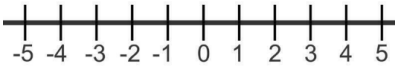
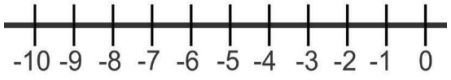
$$3(x+2) \leq 30 \text{ and } 4x+3 > 21$$

Solving Single-Ended Inequalities with Fractions

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	***Example 10 – I do…***	
	$\frac{3x-4}{2} \geq 4$	Represent the solution on the number line below: 
	Example 11 - We do…	
	$\frac{5(x+7)}{3} \leq 10$	List the largest five integers that satisfy this inequality.
	Example 12 - You do…	
	$\frac{2x-2}{5} \leq 10$	List the largest five integers that satisfy this inequality.

Independent Practice

Question 11. Solve and **represent the solutions on the number lines below:**

a) $\frac{2x+1}{3} \geq 5$	b) $\frac{x-15}{4} \leq 3$	c) $\frac{3x+4}{2} \leq -7$
 	 	

Question 12. Solve the inequalities, then list the first five smallest or largest integers that satisfy them:

d) $\frac{3x+3}{2} \geq 21$	e) $\frac{7x-11}{13} \geq 4$	f) $\frac{3x+4}{10} \geq 4$
g) $\frac{3(x+2)}{18} > 8$	h) $\frac{2(3x+1)}{5} \leq 10$	i) $\frac{5(3x+1)}{20} < 4$
j) $\frac{8x+3}{5} \leq -1$	k) $\frac{7x-4}{3} > -6$	l) $\frac{7x-11}{13} < -3$

Solving Single-Ended Inequalities with Negative Coefficients

Steps	*Example 13 – I do…*	*Example 14 - We do…*	*Example 15 - You do…*
1. Where the coefficient of x is negative, we need to add this to both sides of the equation to give 'a positive x' 2. Solve as normal! 3. Isolate x by performing the inverse operation to both sides of the inequality. The solution tells you exactly what x is greater or smaller than.	Solve:		
	$10 - x < 3$	$\frac{25 - x}{2} \geq -9$	$50 \leq 2(5 - 2x)$

Independent Practice

Question 6. Solve each of the inequalities. Show all your working.

a) $24 - x > 17$	b) $-n + 4.3 \leq 3.1$	c) $3 \geq -10 - y$	d) $31 - 3k > 4$
e) $13 < -8 - 3a$	f) $-\frac{x}{2} \geq 12.3$	g) $5(3 - 2x) \leq -35$	h) $\frac{50-x}{7} \geq 5$

Solving Single-Ended Inequalities with Unknowns on Both Sides

	Example 16 – I do…	*Example 17 - We do…*	*Example 18 - You do…*
Steps	**Solve:**	**Solve:**	**Solve:**
1.Identify the coefficient with the lowest value and eliminate this coefficient using the inverse 2. Use the inverse operation to solve the equation	$6x - 4 \leq 4x + 10$	$3x + 12 \geq 7x + 4$	$10x + 3 > 3x + 31$

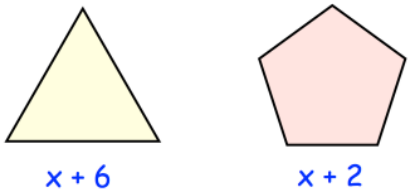
Independent Practice

Question 9. Solve the inequalities below:

a) $8x + 5 > 2x + 23$	b) $7x + 10 \geq 2x + 40$	c) $9x - 6 > 7x + 4$
d) $10x - 3 \leq 4x + 21$	e) $13x - 7 < 3x + 73$	f) $14x + 3 > 4x + 28$

g) $2x + 5 < 10x - 35$	h) $5x + 6 \geq 7x + 2$	i) $x - 6 \leq 10x + 3$
j) $5x + 2 > 10 - 3x$	k) $7x + 20 \leq 40 - 3x$	l) $3x - 6 > 9 - 2x$
m) $x - 3 \leq 9 - 2x$	n) $13x - 54 > 6 - 7x$	o) $x + 3 \leq -2 - 4x$
p) $5x - 24 > 6 - 7x$	q) $4x - 2 \geq -6 - 4x$	r) $10x - 80 > 12 - 6x$

Question 10. The perimeter of the regular pentagon is larger than the perimeter of the equilateral triangle.



a) Form an inequality in terms of x.

b) Solve the inequality to find the possible range of values for x.

Question 11. Megan bought seven crates of apple juice and Adil bought four crates of apple juice. Each crate contained the same number of bottles of apple juice.

When Megan gave Adil ten bottles of juice, Adil then had more bottles than Megan.

a) Form an inequality to represent this scenario, using x to represent crates.

b) Find the maximum number of bottles of apple juice in a crate.

Skills Check

$x + 8 \geq 14$

a) $x \geq 22.$ b) $x \geq 6$ c) $x \geq 1.75.$ d) $x \geq 112$

Solve: $2x - 5 \geq 13$

a) $x \geq 9$ b) $x \geq 4$ c) $x \geq 18.$ d) $x \geq 8$

Solve $3x - 2 \geq -11$

a) $x \geq 3$ b) $x \geq -4$ c) $x \geq 4$ d) $x \geq -3$

Solve: $2(4x - 3) > 10$

a) $x > 10.$ b) $x > 6$ c) $x > 5$ d) $x > 2$

Solve: $\frac{2x+10}{4} < 8$

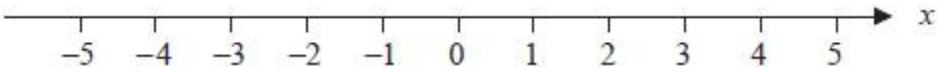
a) $x < 11$ b) $x < -4$ c) $x < 4$ d) $x < 10$

Solve: $8x + 5 > 6x + 7$

a) $x > 6.$ b) $x > 1$ c) $x > 4$ d) $x > 10$

Question 1.
Here is a number line.

(a) On this number line, show the inequality $-2 \leq x < 3$



(2)

(b) Solve $5n + 3 > 27$

.....

(2)

(Total for question = 4 marks)

Question 2. Solve $7x - 27 < 8$

.....

(Total for question = 2 marks)

Question 3. Solve

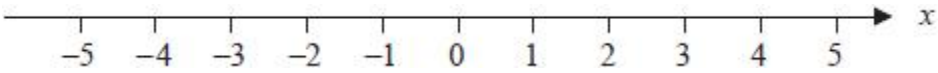
$$\frac{5x}{2} + 3 > 18$$

.....

(Total for question = 3 marks)

Question 4.

(a) On the number line, show the inequality $x < 4$



(2)

$3 < y \leq 7$ where y is an integer.

(b) Write down all the possible values of y.

.....

(2)

(c) Solve $3x + 5 \geq x + 17$

.....

(3)

(Total for question = 7 marks)

Question 5. Solve

$$x + 11 \leq 5 - \frac{1}{2}x$$

.....

(Total for question = 3 marks)

Subtopic 2D – Solving Double-Sided Inequalities

LEARNING OUTCOME: To understand and solve double inequalities

Objectives:

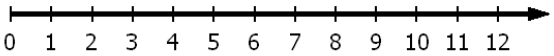
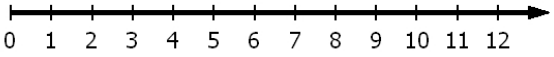
1. To be able to solve double inequalities where the inequality doesn't need to be split.
2. To be able to solve double inequalities where the inequality needs to be split.

Keywords:
- Inequality
- Solution
- Unknown
- Variable

Introduction – read and annotate:

1. Now that you've practised solving single inequalities, it's time to extend those skills to double
2. inequalities. A double inequality compares three expressions at once and shows how a variable fits
3. between two values. Sometimes you can solve it just like a single inequality by working on all parts at
4. once, while other times you may need to split it into two separate inequalities. We'll look at both
5. types so you can see when each method is needed.

Solving Double Inequalities with No Splitting

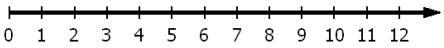
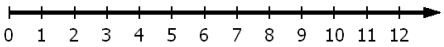
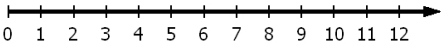
Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to **all three parts** of the inequality.	***Example 1 – I do…***	
	$8 < x + 7 \leq 10$	Represent the solution on the number line below: 
	Example 2 - We do…	
	$6 < 3x \leq 18$	List the integers that satisfy this inequality.
	Example 3 - You do…	
	$2 < x - 3 < 5$	Represent the solution on the number line below: 

Steps	Solve:	
1) Recall your steps required for solving equations. 2) Isolate x by performing the inverse operation to **all three parts** of the inequality.	***Example 4 – I do…***	
	$3 \leq 2x - 3 < 7$	Represent the solution on the number line below: 0 1 2 3 4 5 6 7 8 9 10 11 12
	Example 5 - We do…	
	$2 < \frac{x}{3} + 7 < 10$	List the integers that satisfy this inequality.
	Example 6 - You do…	
	$6 < 3(x + 4) < 12$	List the integers that satisfy this inequality.

Independent Practice

Question 1. Solve the inequalities and showthe solution on the number line.

a) $4 < 2x < 10$	b) $6 < 3x < 15$	c) $4 < x + 1 < 10$
0 1 2 3 4 5 6 7 8 9 10 11 12	0 1 2 3 4 5 6 7 8 9 10 11 12	0 1 2 3 4 5 6 7 8 9 10 11 12

d) $0 < \frac{x}{3} < 3$	e) $4 < x - 2 < 10$	f) $4 < x + 2 < 10$
		

Question 2. Solve the inequalities and list the integers that satisfy the inequality.

a)$4 \leq 2x < 5$	b)$-4 \leq 2x < 5$	c) $4 < 2x - 1 < 10$
d) $4 < 3x - 2 < 10$	e) $2 < \frac{x}{3} - 2 < 5$	f) $-6 < 3(x - 2) < 12$
g) $-4 < x - 2 < 10$	h) $-1 \leq \frac{x-1}{2} \leq 2$	i) $-10 \leq 2x - 1 \leq -4$

Question 3. Solve the inequalities below.

a) $-10 \leq 4x - 2 \leq -4$	b) $-10 \leq 4x - 2 \leq 4$	c) $-10 \leq 4x - 2 \leq 8$
d) $-20 < \frac{1}{4}x \leq 8$	e) $-20 < \frac{1}{2}x \leq 8$	f) $-20 < \frac{1}{2}x - 8 \leq 8$
g) $-20 < \frac{x-8}{2} \leq 8$	h) $-20 < \frac{8+x}{2} < 8$	i) $-20 < \frac{2(x+1)}{8} < 8$

Question 4. In each question, explain what mistake has been made and correct the answer.

a)	b)
$7 < 3x + 4 < 10$ $3 < 3x < 10$ $1 < x < 10$	$1 < \frac{3x+5}{2} \leq 5$ $2 < 3x + 5 < 10$ $-3 < 3x < 5$ $-1 < x < \frac{5}{3}$

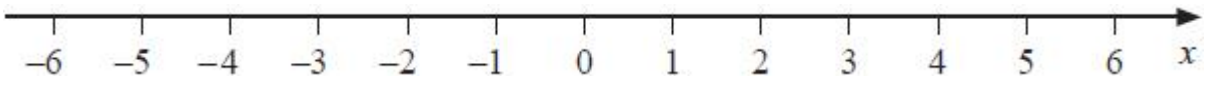
Exam Question

(a) Solve $14n > 11n + 6$

.....

(2)

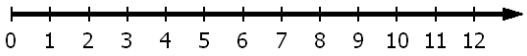
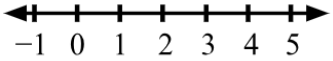
(b) On the number line below, show the set of values of x for which $-2 < x + 3 \leq 4$



(3)

(Total for question = 5 marks)

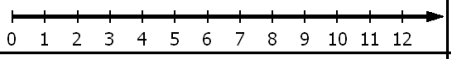
Solving Double-Sided Inequalities Where a Split is Needed

Steps	Solve:
1) Split the double inequality into two separate inequalities. 2) Solve each inequality separately. 3) Write your two separate answers as one double inequality.	***Example 7 – I do…***
	$3 - x \leq 2 < 10 - 2x$
	Represent the solution on the number line: 
	Example 8 - We do…
	$x + 7 \leq 6x - 3 < 5x + 18$
	List the largest and smallest integers that satisfy this inequality.
	Example 9 - You do…
	$1 + x < 5 \leq 7 + 2x$
	Represent the solution on the number line: 

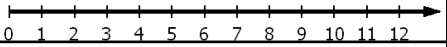
Independent Practice

Question 1. Solve the inequalities and showthe solution on the number line.

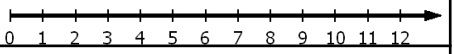
a) $x + 1 < 5 < 2x + 3$



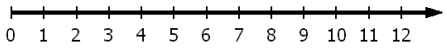
b) $3x - 2 < 7 \leq x + 1$



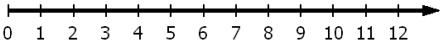
c) $x + 3 < 5 < 2x + 3$



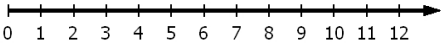
d) $x + 2 < 7 \leq 3x + 2$



e)$3(4 + x) > -6 \geq x - 2$



f)$2x + 1 \leq -7 < \frac{x}{3} + 4$



Question 2. Solve the inequalities and list the integers that satisfy the inequality.

g) $12 + x \geq 10 > 2x - 1$	h) $4x - 15 \leq 7 < 3x + 1$	i) $x - 6 < -2 \leq 2x + 1$
j) $2(x + 3) \leq 10 < 5x + 1$	k) $6 + x < 8 \leq 4(x - 2)$	$x - 2 \geq 4 > \frac{2x}{5} + 1$

Question 2. Solve the inequalities and list the integers that satisfy the inequality.

p) $7x - 1 \leq 10x + 2 \leq 3x + 23$	q) $5x - 14 \leq 3(x + 4) \leq 4x$
r) $6x - 10 \leq 2(x + 5) \leq 5x$	s) $2x - 14 < 7 - 5x \leq -3x + 11$
t) $2(4x - 3) < 5(x + 3) < 2(3x + 5) - 1$	u) $\frac{2x-5}{2} < 3x + 2 < \frac{x+20}{3}$

Question 4.

$$2x + 1 \leq 3x \leq 2x + 17$$

a) Does the following inequality need to be split into two to solve, or is it possible to solve it as one, like the first two example sets? You must explain your answer.

b) Hence, solve the inequality.

Skills Check

In which of the following inequalities could $\boldsymbol{d} = \mathbf{2}$?

Inequality 1

$$-5 < 1 - 3d \leq 0$$

Inequality 2

$$-1 < \frac{1-d}{3} \leq 0$$

A Only Inequality 1

B Only Inequality 2

C Both Inequalities

D Neither

$$-12 < 2x + 4 < 24$$

What is the solution to this inequality?

A $-8 < x < 10$

B $8 < x < 10$

C $-20 < x < 52$

D $-10 < x < 8$

What is the solution to this inequality?

$$-10 \leq \frac{2x-8}{3} < 4$$

A $-19 \leq x < 10$

B $-11 < x < 10$

C $-11 \leq x < 10$

D $-19 < x \leq 2$

Subtopic 3 – Rearranging Formulae

LEARNING OUTCOME: To be able to rearrange formulae

Objectives:

1. To be able to rearrange formulae with one-step
2. To be able to rearrange formulae with two-steps
3. To be able to rearrange formulae requiring many-steps

Keywords:
- Equation
- Subject
- Inverse
- Formulae

Introduction

1. Rearranging formulae is a lot like solving equations—you still use inverse operations on both sides to
2. keep things balanced.
3. The difference is, instead of finding a number, you're working to make a **specific variable standalone**.

4. We call this variable the **subject** of the equation – the single variable that everything else is
5. equal to. The subject should have **no coefficient, constant, power or other function attached to it**,
6. and **appear only once** in the formula, usually isolated on the left.

Examples:

$A = \pi r^2$

$y = mx + c$

$P = 2l + 2w$

$2n + 3 = T$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Non-examples:

$x = 3y + 2x$

$5g = 3h - 7$

$m + 3 = n^2 + 2\mathrm{p}$

$a^2 + b^2 = c^2$

7. Making a variable the subject can also be described as rewriting the equation to express that variable
8. in terms of the others.

Is x the subject of the formula?

Formula	Yes/ No	Formula	Yes/ No
a) $x = y + 5$		j) $y + 5 = -x$	
b) $y + 5 = x$		k) $\frac{1}{x} = z + 8$	
c) $x + 5 = y$		l) $x^2 = y + 3$	
d) $x + y = a$		m) $x = y^2 + 3x$	
e) $xy = b$		n) $2x = z^2 + y$	
f) $x = yz - 8$		o) $y^x + 7 = a$	
g) $x = yz - b$		p) $\sqrt{x} = y^2 + 3$	
h) $x = yz - x$		q) $x = \sqrt{\frac{y^2+3}{2}}$	
i) $x = yz - x^2$		r) $\sqrt{\frac{y^2+3}{2}} = x$	

One-Step Rearranging

One-step Rearranging (Addition/ Subtraction)

	Example 1. I do…	*Example 2. We do…*	*Example 3. You do…*
Steps	Make a the subject of the following:	Make x the subject of the following:	Make y the subject of the following:
1) Circle the variable you want to make the subject. 2) Use the inverse operation to isolate the variable.	$a + 5 = x$	$c = x - b$	$y + 3a - 6 = x$

Independent Practice

Question 1. Make x the subject of the following:

a) $x + p = k$	b) $x - 2p = k$	c) $x - 2p = k + t$	d)) $4a = x - k$
e) $x + y = 7$	f) $x + y = 12$	g) $x - y = 12$	h) $x - 2y^2 = 12$
i) $12 = x - 2y + z$	j) $12 + 3y = x + 2y$	k) $-2y = x - 12 - 3z$	l) $\frac{a-b}{2} = x + c$

One-step Rearranging (Multiplication/ Division)

	Example 4. I do…	*Example 5. We do…*	*Example 6. You do…*
Steps	Make r the subject of the following:	Make y the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use the inverse operation to isolate the variable.	$p = \frac{r}{8}$	$py = a - b$	$-3bx = c$

Independent Practice

Question 2. Make x the subject of the following:

a) $2x = b$	b) $\frac{x}{2} = b$	c) $\frac{x}{c} = b$	d) $xc = b$
e) $ab = cx$	f) $ab = \frac{x}{d}$	g) $y = \frac{x}{bc}$	h) $mx = w + q$
i) $6cx = a - b$	j) $5xz = b - c$	k) $\frac{x}{2m} = p - t$	l) $y = -bcx$
m) $y = -\frac{x}{a}$	n) $y = -\frac{x}{\sqrt{a}}$	o) $y = -\frac{x}{\sqrt{bc}}$	p) $-\frac{x}{2m} = p - t$

	Example 7. I do...	***Example 8. We do...***	***Example 9. You do...***
Steps	Make x the subject		
Circle the subject that we want to isolate.	$x(a + b) = c$	$\frac{x}{y - c} = b$	$x(a + b - c) = m$
Identify what needs to be eliminated.			
Inverse the operation, and apply to both sides.			

Independent Practice

Question 3. Make x the subject of the formulae:

a) $\frac{x}{a+c} = y$	b) $\frac{x}{a-c} = y$	c) $x(a + 5) = y$	d) $\frac{x}{c+a} = m$
e) $m = \frac{x}{a+b}$	f) $x(a - c) = y$	g) $x(a - c) = y + b$	h) $\frac{x}{a-b+c} = d$

	Example 10. I do…	*Example 11. We do…*	*Example 12. You do…*
Steps	Make x the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use the inverse operation to isolate the variable.	$\sqrt{x} = y$	$x^2 = y + 4$	$x^2 = z$

Independent Practice

Question 4. Make x the subject of the following:

a) $x^2 = y$	b) $y = x^2$	c) $\sqrt{x} = a$	d) $x^2 = a + 5$
e) $x^2 = 2y$	f) $\sqrt{x} = y + 1$	g) $\sqrt{x} = y^2 + 1$	h) $x^2 = y^2 + 1$
i) $x^3 = m$	j) $x^3 = m + a$	k) $\sqrt[3]{x} = b$	l) $\sqrt[3]{x} = b + \mathrm{a}$

Mixed Practice

Question 5.
Given that $3y = 2x$
a) Write y in terms of x.

b) Write x in terms of y.

Question 6.
The area of a circle is given as $A = \pi r^2$.

Make π the subject of the formula.

Question 7.
Jean is attempting to make g the subject of the formula

$$g^2 = j - 5$$

Here is her answer:

$$g = \sqrt{j} - 5$$

What mistake has she made? What is the correct answer?

	Example 13. I do…	Example 14. We do…	Example 15. You do…
Steps	Make x the subject		
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable.	$y = \frac{3x}{5}$	$y = \frac{4x}{b}$	$y = \frac{ax}{b}$

Independent Practice

Question 8. Make x the subject of the following:

a) $y = \frac{2x}{5}$	b) $m = \frac{2x}{a}$	c) $\frac{ax}{6} = b$	d) $\frac{bx}{a} = 10$
e) $y = \frac{2x}{c}$	f) $m = \frac{bx}{a}$	g) $\frac{ax}{6} = -b$	h) $\frac{bx}{a} = -10$
i) $y = \frac{2x}{3c}$	j) $3m = \frac{2x}{a}$	k) $\frac{ax}{6} = 2b$	l) $\frac{7bx}{a} = -10$

Two-Step Rearranging

Two-step Rearranging (addition/ subtraction with division)

	Example 16. I do…	*Example 17. We do…*	*Example 18. You do…*
Steps	Make x the subject of the following:	Make x the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operation<u>s</u> to isolate the variable.	$2x - a = y$	$y = abx - c$	$y = mx + c$

Independent Practice

Question 9. Make x the subject of the following:

a) $y = 4x + 3$	b)$y = 4x - 3$	c) $y = 4x + a$	d) $y = ax - b$
a) $y = ax + b$	b) $y = bx - a$	c) $y = ax - bc$	d) $y = tx - r$
e) $y = ax + bcd$	f) $b^2cd + ax = y$	g)) $2x + 2y = c$	h) $ax + by = c^2$

1. Where there are brackets in the formula, there are two approaches to change the subject:
2. (a) Expand the brackets and rearrange as previously
3. (b) Consider the multiplication of the bracket as an operation to be 'undone' first.

	Example 19. I do…	
Steps	Make x the subject of $y = 4(x+3)$	
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable, using method (a) or (b)	***Method (a)***	***Method (b)***
	Example 20. We do…	***Example 21. You do…***
	Make x the subject of the following:	Make x the subject of the following:
	$a(x-5) = y$	$y = m(x+a)$

Independent Practice

Question 10. Make x the subject of the following:

a) $y = 5(x+3)$	b) $y = 5(x-3)$	c) $y = 5(x+a)$	d) $y = a(x-b)$
a) $y = a(x+b)$	b) $y = b(x-a^2)$	c) $y = a(x-bc)$	d) $y = b(cd+ax)$

Carefully consider the order of operations for divisions across single and multiple terms.

	Example 22. I do…	***Example 23. We do…***	***Example 24. You do…***
Steps	Make y the subject of the following:	Make x the subject of the following:	Make d the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable.	$\frac{y+3}{5}=z$	$y=\frac{x-a}{2}$	$\frac{d+e}{4}=f$

Independent Practice

Question 11. Make x the subject of the following:

a) $\frac{x+4}{3}=m$	b) $\frac{x+3}{5}=h$	c) $t=\frac{x-v}{4}$	d) $\frac{x+b}{2a}=r$
e) $\frac{x+3}{c}=h$	f) $\frac{w+x}{u}=3z$	g) $\frac{x+3t}{m}=2c$	h) $\frac{x+q}{p}=p$

	Example 25. I do…	***Example 26. We do…***	***Example 27. You do…***
Steps	Make x the subject		
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable.	$\frac{x}{5}+3=z$	$\frac{x}{2}-a=y$	$\frac{x}{4}+e=f$

Independent Practice

Question 12. Make x the subject of the following:

a) $h = \frac{x}{2} - f$	b) $\frac{x}{6} - 5 = w$	c) $t = \frac{x}{4} + 4r$	d) $5y = \frac{x}{3} - 7z$
e) $\frac{x}{n} + 2 = w$	f) $\frac{x}{t} - e = 5k$	g) $d = \frac{x}{b} + c$	h) $g = 7 + \frac{5x}{3}$

Mixed Practice

Question 13. Make x the subject

a) $\frac{x-5}{2} = y$	b) $y = \frac{x}{2} - 5$	c) $2x - 5 = y$	d) $y = \frac{1}{2}x - 5$
e) $\frac{x}{a} + 2b = y$	f) $\frac{b+x}{2a} = y$	g) $y = a(x + 2b)$	h) $b + \frac{x}{2a} = y$

Question 14.

Pritesh has tried to rearrange $a = \frac{x}{4} + b$ to make x the subject.

He has made a mistake – can you explain it?

What is the correct rearrangement?

$$a = \frac{x}{4} + b$$

$$4a = x + b$$

$$4a - b = x$$

	Example 28. I do…	*Example 29. We do…*	*Example 30. You do…*
Steps	Make x the subject of the following:	Make y the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable.	$7 + x^2 = y$	$\frac{\sqrt{y}}{a} = b$	$5x^3 = p$

Independent Practice

Question 15. Make x the subject of the following:

a) $x^3 - 4 = y$	b) $y = x^2 + c$	c) $y = x^3 + c$	d) $(x + c)^3 = y$
e) $\sqrt{x} + 6 = y$	f) $\sqrt{x} + k = y$	g) $\sqrt[3]{x} - k = y$	h) $\sqrt{x - k} = y$
i) $y = 3x^2$	j) $y = \sqrt{3x}$	k) $\sqrt{\frac{x}{7}} = y$	l) $\left(\frac{x}{5}\right)^3 = y$

Skills Check

Rearrange this formula to make p the subject:

$$m = 5p$$

A $p = \frac{5}{m}$

B $p = m - 5$

C $p = 5m$

D $p = \frac{m}{5}$

Rearrange this formula to make r the subject:

$$g = 2r + 3$$

A $r = \frac{g+3}{2}$

B $r = g + 3 \div 2$

C $r = \frac{g-3}{2}$

D $r = g - \frac{3}{2}$

Make x the subject of the formula

$$f = \sqrt{x+3}$$

A $x = 2f - 3$

B $f^2 = x + 3$

C $x = f^2 - 9$

D $x = f^2 - 3$

Tom and Katie are discussing rearranging $t = \sqrt{u+9}$ to make u the subject

Tom says a correct first line of working could be: $t = \sqrt{u} + 3$

Katie says a correct first line of working could be: $t^2 = u + 9$

Who is correct?

A Only Tom

B Only Katie

C Both Tom and Katie

D Neither is correct

	Example 31. I do…	*Example 32. We do…*	*Example 33. You do…*
Steps	Make x the subject of the following:	Make y the subject of the following:	Make x the subject of the following:
1) Circle the variable you want to make the subject. 2) Use inverse operations to isolate the variable.	$\frac{7x+b}{3}=9$	$2y^2+b=m$	$\frac{\sqrt{x}}{3}+9=\mathrm{r}$

Independent Practice

Question 16. Make x the subject of the following:

a) $\frac{7x+m}{11}=1$	b) $\frac{7x^2+m}{11}=1$	c) $5(x^2-ab)=c$	d) $5x^2+ab=c$
e) $7x+c-b=9$	f) $\frac{p-4+2x}{m}=m$	g) $\frac{3(p-4+2x)}{m}=n$	h) $\frac{x^2}{3}-g=a$
i) $\frac{\sqrt{x}}{g}+b=m$	j) $\frac{\sqrt[3]{x}}{a}-d=f$	k) $\frac{x^3}{2}=g+5$	l) $\frac{3x^2}{10}+b=a^2$
m) $\frac{x^2}{5}+ab=a^2$	n) $\frac{x^3}{5}+ab=a^2$	o) $(m-3)x^2=g$	p) $(m-n)x^2=g$

	Example 34. I do…	*Example 35. We do*	*Example 36. You do*
Steps	Make x the subject of the following:	Make a the subject of the following:	Make h the subject of the following:
1) Circle the variable you want to make the subject & consider the coefficient carefully. 2) Where the coefficient is negative, we need to add this to both sides of the equation to give 'a positive' subject. 3) Rearrange as normal by isolating the variable using inverse operations.	$y = 5 - x$	$\mathrm{b} - 4a = 9$	$5g - h^2 = 3j$

Independent Practice

Question 17. Make x the subject of the following:

a) $4a - x = 7$	b) $4a - x = 7b$	c) $a - 3x = 7$	d) $-3x + 4a = 7$
e) $5z - x^2 = y^2 + 1$	f) $y = 12 - \sqrt{x}$	g) $5b - \frac{x}{4} = c$	h) $\frac{a-x}{7} = b$

Challenge

Question 18. Make x the subject of the following:

a) $5x + 7y = 10 - x$	b) $4x - 5 = 3x + 6y$

Question 1.

Make h the subject of the formula $x = 5h + 8$

.....

(Total for Question is 2 marks)

Question 2.

Make h the subject of the formula $t = \frac{gh}{10}$

h = ...

(Total for Question is 2 marks)

Question 3.

Make p the subject of this formula.

$$q = \frac{p}{r} + s$$

.....

(Total for question = 2 marks)

Question 4.

Make *g* the subject of the formula

$$T = \sqrt{\frac{g + 6}{2}}$$

.....

(Total for question = 3 marks)

Question 5.

$$T = 4v + 3$$

(a) Work out the value of T when $v = 2$

T = ...

(2)

(b) Make v the subject of the formula $T = 4v + 3$

.....

(2)

(Total for question = 4 marks)

Question 6.

$$v^2 = u^2 + 2as$$

$u = 12 \quad a = -3 \quad s = 18$

(a) Work out a value of v.

.....

(2)

(b) Make s the subject of $v^2 = u^2 + 2as$

.....

(2)

(Total for question = 4 marks)